

# WASP-135b

R-0187

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-135\_250712 · created 2026-07-16T20:19:42+00:00

## BENCHMARK: INACCURATE

### Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2460868.7152 +/- 0.0042 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.176 +/- 0.017                 |
| Transit depth (Rp/Rs)^2                           | 3.08 +/- 0.61 [%]               |
| Orbital Inclination (inc)                         | 82.6 +/- 1.57                   |
| Airmass coefficient 1 (a1)                        | 1.139 +/- 0.032                 |
| Airmass coefficient 2 (a2)                        | -0.13 +/- 0.11                  |
| Scatter in the residuals of the lightcurve fit is | 2.81 %                          |
| Best Comparison Star                              | #3 - [328, 275]                 |
| Optimal Aperture                                  | 2.52                            |
| Optimal Annulus                                   | 15.35                           |
| Transit Duration (day)                            | 0.075 +/- 0.011                 |

### Accuracy vs published ephemeris (ExoClock/NEA)

|                                      |                                                      |
|--------------------------------------|------------------------------------------------------|
| Rp/R $\blacksquare$ fit vs published | 0.176 $\pm$ 0.017 vs 0.139 (deviation 2.1 $\sigma$ ) |
| Duration fit vs published            | 0.075 d vs 0.0687 d (deviation 0.55 $\sigma$ )       |
| Mid-time O-C                         | -31.7 min                                            |
| Depth SNR                            | 10.0                                                 |
| Residual scatter                     | 2.81 %                                               |

### Light curve

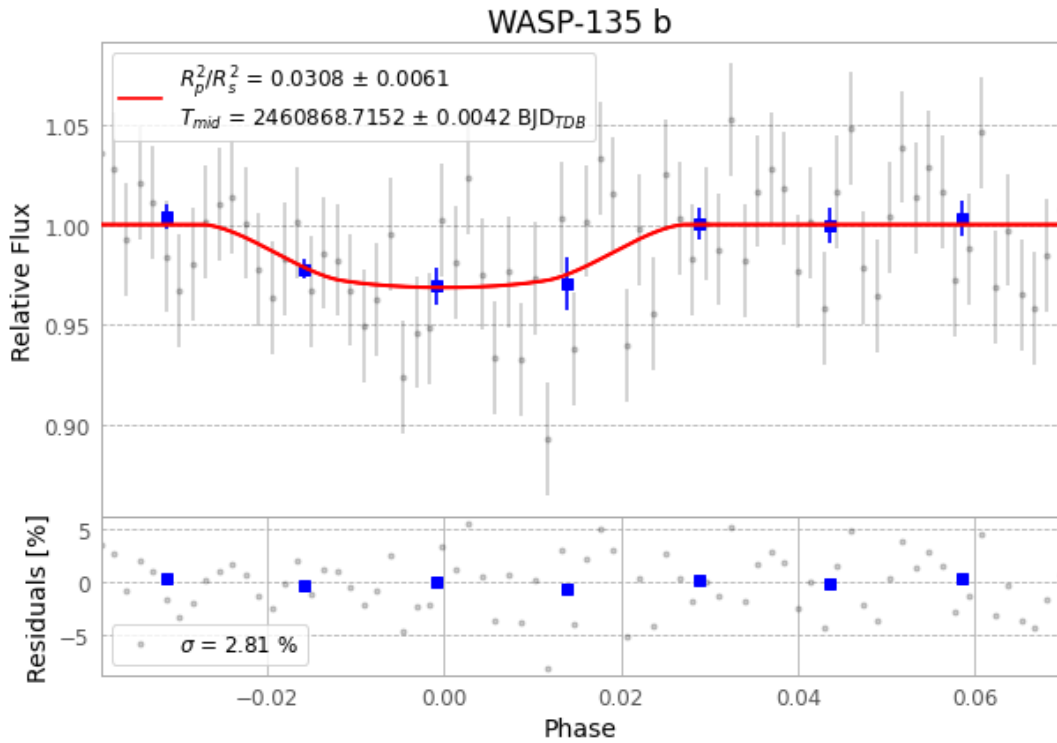


Figure 1 — FinalLightCurve\_WASP-135 b\_2025-07-11.png (SHA-256 436c85805f130b0b...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [356, 196], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 03c2dbca15088635...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 378b18d

## Data provenance

74 FITS frames (49 MB), WASP-135250712034515.FITS → WASP-135250712072415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-135-250712.log (8521553bf3e2...), FinalLightCurve\_WASP-135 b\_2025-07-11.png (436c85805f13...), FinalLightCurve\_WASP-135 b\_2025-07-11.csv (3efe5dcd7812...)

## Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

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Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 20:19Z.